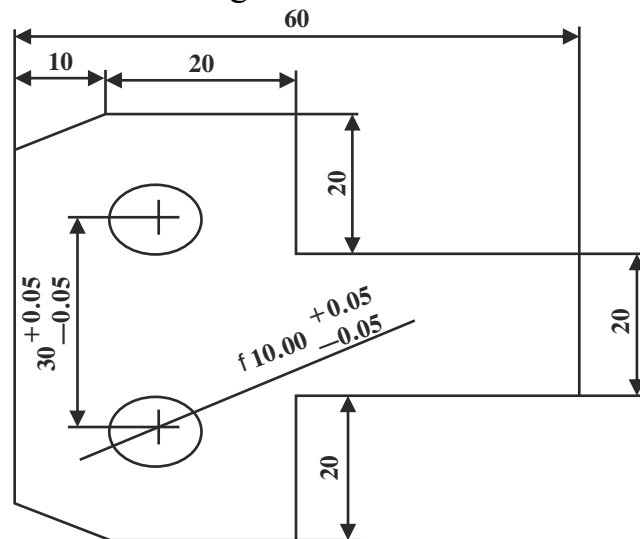


Q.24 Design and draw a die to produce sheet metal component as shown in fig 'A' of thickness 2 mm. The Component is made of Brass having an ultimate shear strength of 350N/mm^2 . Your design should include:

1. At least two views of the die
2. Economic strip layout
3. Bill of materials
4. Press tonnage calculations



- Dimensions units may be taken in mm.
- Assume suitable dimensions if not specified or clearly visible in the figure.

Fig 'A'

- Q.25 a) Differentiate between compound die and combination die with neat sketch.
b) Differentiate between coining and embossing with the help of neat sketch.

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**3rd Sem / Mechanical (Tool & Die Design),
CNC, CAD/CAM**

Subject : Press Tool Design and Drawing

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 What should be the appropriate thickness of the metal sheet when it is used as a raw material for the sheet metal operations?

- a) 1 cm to 5 cm b) 5 cm to 10 cm
c) 7.5 mm to 15 mm d) 0.4 mm to 6 mm

Q.2 Bevelling is particularly suitable for shearing of?

- a) Thin blanks b) Thick blanks
c) Very thin blanks d) Medium thin blanks

Q.3 The cutting clearance for CRCA sheet in press tool is _____

- a) 2% b) 8%
c) 15% d) 20%

Q.4 Punch holder is located and clamped on

- a) Bottom plate b) Top plate
c) Valve d) Compressor

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- Q.5 In blanking tool, the punch is made up of
 a) HC-HCr steel b) Cast iron
 c) Mild steel d) All of above
- Q.6 In die cutting operation, punching a number of holes in a sheet is called
 a) Parting b) Notching
 c) Perforating d) None

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define shearing force.
 Q.8 What is idle station.
 Q.9 Define spring back action.
 Q.10 Give the application of coining process.
 Q.11 Define die cushioning.
 Q.12 Define shut height

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Differentiate between bending and curling.
 Q.14 Explain feeding mechanism with diagram.
 Q.15 Explain the principle of grain direction and its importance with diagram.
 Q.16 Explain the geometry of punching tool with diagram.
 Q.17 Write short note on material utilization.
 Q.18 Classify press tools.

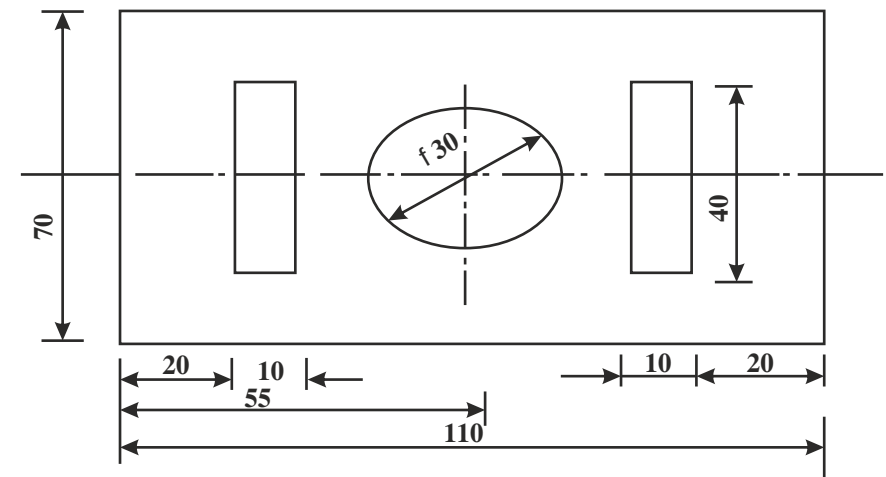
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- Q.19 Explain the difference between noching and shaving with diagram.
 Q.20 Calculate the force required to cut notch of 5x5mm out of a sheet of 50x50mm and 2mm thickness, when the shear stress is 300 N/mm²
 Q.21 Give a specification of any bending press.
 Q.22 Describe the BIS standards for bottom & top pillars.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Sketch & Design a progressive die to make a steel stamping from 2mm thick metallic sheet. The ultimate shear strength of the brass material is 320 N/mm².



- Dimensions units may be taken in mm.
- Assume suitable dimensions if not specified or clearly visible

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